

Git & GitHub Command Reference

A quick-reference cheat sheet of commonly used Git and GitHub CLI commands.

1. Setup & Configuration

Command	Description
<code>git init</code>	Initialize a new local Git repository
<code>git clone <url></code>	Copy a remote repository to your local machine
<code>git config --global user.name "name"</code>	Set your Git username globally
<code>git config --global user.email "email"</code>	Set your Git email globally
<code>git config --list</code>	Show all current Git configuration settings

2. Basic Workflow

Command	Description
<code>git status</code>	Show changed/staged/untracked files in working directory
<code>git add <file></code>	Stage a specific file for the next commit
<code>git add .</code>	Stage all changed files in the current directory
<code>git commit -m "message"</code>	Save staged changes as a new commit with a message
<code>git commit -am "message"</code>	Stage all tracked changes and commit in one step
<code>git diff</code>	Show unstaged changes between working directory and last commit
<code>git diff --staged</code>	Show changes that are staged but not yet committed
<code>git log</code>	Show commit history for the current branch
<code>git log --oneline</code>	Show condensed one-line-per-commit history
<code>git show <commit></code>	Show details and changes of a specific commit

3. Branching

Command	Description
<code>git branch</code>	List all local branches
<code>git branch <name></code>	Create a new branch
<code>git branch -d <name></code>	Delete a local branch
<code>git checkout <branch></code>	Switch to an existing branch
<code>git checkout -b <branch></code>	Create and switch to a new branch
<code>git switch <branch></code>	Switch to an existing branch (modern alternative)
<code>git switch -c <branch></code>	Create and switch to a new branch (modern alternative)
<code>git merge <branch></code>	Merge the specified branch into the current branch
<code>git rebase <branch></code>	Reapply current branch's commits on top of another branch
<code>git branch -a</code>	List all local and remote branches

4. Remote Repositories (GitHub)

Command	Description
git remote -v	List remote repositories linked to this repo
git remote add origin <url>	Link a remote repository named 'origin'
git fetch	Download objects/refs from remote without merging
git pull	Fetch and merge changes from the remote branch
git pull --rebase	Fetch remote changes and rebase local commits on top
git push	Upload local commits to the remote repository
git push -u origin <branch>	Push a branch and set it to track the remote branch
git push origin --delete <branch>	Delete a branch on the remote repository

5. Undoing Changes & Stashing

Command	Description
git reset <file>	Unstage a file, keeping changes in working directory
git reset --hard <commit>	Discard all changes and reset to a specific commit
git reset --soft <commit>	Move HEAD to a commit, keeping changes staged
git revert <commit>	Create a new commit that undoes a previous commit
git checkout -- <file>	Discard local changes in a specific file
git clean -fd	Remove untracked files and directories from working dir
git stash	Temporarily save uncommitted changes
git stash pop	Reapply the most recently stashed changes and remove stash
git stash list	List all stashed change sets
git cherry-pick <commit>	Apply a specific commit from another branch onto current

6. Inspecting & Tagging

Command	Description
git blame <file>	Show who last modified each line of a file
git tag <name>	Create a tag at the current commit (e.g. for releases)
git tag -a <name> -m "msg"	Create an annotated tag with a message
git push origin --tags	Push all local tags to the remote repository
git show-branch	Show branches and their commits in a visual summary
git reflog	Show a log of where HEAD has pointed, useful for recovery

7. GitHub CLI (gh)

Command	Description
gh auth login	Authenticate the GitHub CLI with your GitHub account
gh repo create <name>	Create a new repository on GitHub
gh repo clone <owner/repo>	Clone a GitHub repository using the CLI
gh repo view --web	Open the current repository in a web browser

Command	Description
gh issue list	List issues in the current repository
gh issue create	Create a new issue interactively
gh pr create	Create a new pull request from the current branch
gh pr list	List open pull requests
gh pr checkout <number>	Check out a pull request locally by its number
gh pr merge <number>	Merge a specified pull request
gh pr view --web	Open the current pull request in a web browser
gh release create <tag>	Create a new GitHub release